Giammarlo Adrian S Abad BSIT 2-1N Object Oriented Programming Act#1 Variables

PART A - VARIABLE IDENTIFICATION TABLE

Variable Name	Data Type	Variable Type	Stored In
name	String	Instance	Heap Memory
grade	int	Instance	Heap Memory
s1	Object	Local	Stack Memory
s2	Object	Local	Stack Memory

PART B - EXPLANATION

Statement	What Happend	Memory Impact
Student s1 = new Student("John", 85);	A new Student Object has been instatiated, with the properties "John" for its name & 85 for its grade.	The new instance of the Student Object has been added to the heap, while s1, the reference variable to the object is added to the stack memory.
Student s2 = new Student ("Maria", 90)	Another Student has been Instantiated, this time its properties is "Maria" for the name & 90 for the grade.	Another Instance of a Student Object has been added to the heap, while s2, the reference variable to the object is added to the stack memory.
Student s3 = s2;	An object variable with the name "s3" has been initialized with it pointing to the same reference as s2.	s3 is added to the stack memory, nothing changes to the heap as s2 & s3 are just pointing to one instance.
s1.grade = 95;	s1's object property for grade got its value changed to 95.	s1's student Object got modified on the heap
s3.name = "Ana";	s3's object property for name got its value changed to "Ana".	s3 & subsequently s2's student Object is modified on the heap as both point to the same student object.

PART C - OUTPUT

John 95

Ana 90

Ana 90